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## ME 606: Advanced Solid Mechanics

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| Course Code  | ME 606                      |
| Course Name  | Advanced Solid Mechanics    |
| L-T-P-C      | 3-0-0-3                     |
| Intended for | MS/PhD                      |
| Distribution | Elective                    |
| Approval     | 2nd Adhoc Senate (May 2012) |

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### Course Contents

Shear centre and unsymmetrical bending. Beam columns: beams on elastic foundations, curved beams, Rotating discs and thick cylinders, Virtual work; minimum potential energy; Hamilton's principle. Plate theory: formulation by Hamilton's principle: bending and buckling of homogeneous and sandwich plates. Shell theory: introduction to theory of surface; formulation by Hamilton's principle; membrane, bending and buckling analysis of shells of revolution.

### Text and Reference Books

1. Srinath L.S., **Advanced Mechanics of Solids**, Tata McGraw-Hill, 1980
2. Boresi, A.P. and Sidebottom, O.M., **Advanced Mechanics of Materials**, John Wiley, 1993.
3. Timoshenko, S.P. and Goodier, J.B., **Theory of Elasticity**, McGraw-Hill Kogakusha Ltd., 1970.
4. Reddy, J.N., **Theory and Analysis of Elastic Plates and Shells**, 2nd Edition